

Multidimensional imaging model for diagnosing primary open-angle glaucoma using fundus photographs and optical coherence tomography images

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To cite: Qiu J, Lin J, Shi L, et al. Multidimensional imaging model for diagnosing primary open-angle glaucoma using fundus photographs and optical coherence tomography images. *BMJ Open Ophthalmology* 2025;10:e002274. doi:10.1136/bmjophth-2025-002274

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/bmjophth-2025-002274>).

Received 14 April 2025

Accepted 25 September 2025

ABSTRACT

Aims To develop a multidimensional imaging model to detect primary open-angle glaucoma (POAG) by combining fundus photographs with the retinal nerve fibre layer (RNFL) and ganglion cell-inner plexiform layer (GCIPL) thickness maps and deviation maps.

Methods 1054 eyes of 546 patients were included. Images were randomly divided into training, validation and test datasets at 64%, 16% and 20%. Four basic convolutional neural network algorithms were used to develop the multidimensional imaging model by combining fundus photographs, RNFL and GCIPL deviation maps, with RNFL and GCIPL thickness maps. POAG structure damage was defined as RNFL and GCIPL thinning with neuroretinal rim thinning.

Results Multidimensional imaging model obtained an area under the receiver operating characteristic curve (AUC) of 0.970 (95% CI 0.958 to 0.979), which was superior to the fundus photographs model (AUC, 0.945 (95% CI 0.929 to 0.958); $p < 0.05$), RNFL and GCIPL deviation maps model (AUC, 0.931 (95% CI 0.915 to 0.946); $p < 0.001$), RNFL and GCIPL thickness maps model (AUC, 0.958 (95% CI 0.944 to 0.969)) in the test dataset. There was a significant difference between the multidimensional imaging model (AUC, 0.969 (95% CI 0.934 to 0.988)) and three glaucoma ophthalmologists (attending ophthalmologist 1: AUC, 0.847; attending ophthalmologist 2: AUC, 0.842; resident ophthalmologist: AUC, 0.755; $p < 0.001$ for both) in one fold of the test dataset. Furthermore, the overall AUCs were lower in the myopic group than in the non-myopic group. The overall AUC for lesion detection by each model ($p=0.159$) was: low myopia group (AUC, 0.968 (95% CI 0.940 to 0.985)) > moderate myopia group (AUC, 0.966 (95% CI 0.936 to 0.984)) > high myopia group (AUC, 0.958 (95% CI 0.927 to 0.978)). The multidimensional imaging model outperformed unimodal models in diagnosing myopia combined with POAG.

Conclusion The multidimensional imaging model, combining fundus photographs, RNFL and GCIPL deviation maps, with RNFL and GCIPL thickness maps, could perform better than unimodal models. Its potential to improve the diagnostic accuracy of myopia combined with POAG for various myopia levels is promising. This means it could be a valuable tool for assisting with POAG diagnosis in the clinic, offering hope for improved patient care.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Glaucoma, particularly primary open-angle glaucoma (POAG), is the leading cause of irreversible blindness globally, and early detection is critical for preserving optic nerve function, yet current diagnostic methods face challenges, especially in myopic patients, underscoring the need for improved diagnostic techniques.

WHAT THIS STUDY ADDS

⇒ This study presents a high-performance multidimensional imaging model, which integrates fundus photographs, retinal nerve fibre layer (RNFL) and ganglion cell-inner plexiform layer (GCIPL) deviation maps, with RNFL and GCIPL thickness maps to significantly improve the diagnostic accuracy for POAG, especially in myopic patients, thereby enhancing diagnostic capabilities for glaucoma.

HOW THIS STUDY MIGHT AFFECT RESEARCH PRACTICE OR POLICY

⇒ This study may influence research and clinical practice by establishing a robust multimodal deep-learning approach for glaucoma diagnosis, potentially leading to enhanced diagnostic accuracy and earlier detection of POAG, particularly in myopic patients, thereby informing guidelines and policies for eye care management.

INTRODUCTION

Glaucoma is the leading cause of irreversible blindness worldwide.¹ With retinal ganglion cell death and optic nerve fibre loss in glaucoma, there are characteristic changes in the appearance of the optic nerve head and retinal nerve fibre layer (RNFL).²⁻³ Primary open-angle glaucoma (POAG) is probably the most common type.⁴ Nevertheless, the RNFL generally suffers damage before the onset of functional defects.⁵⁻⁷ Therefore, identifying POAG early is extremely important to preserving optic nerve function. In addition, myopic fundus structural changes have increased the difficulty of diagnosing



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patients with POAG.⁸ The number of people with myopia worldwide is expected to reach 4.758 billion by 2050.⁹ Therefore, it is significant to diagnose myopic patients with POAG.

Fundus photography may reveal the presence of neuroretinal rim thinning, RNFL defects, optic disc haemorrhages and parapapillary atrophy (PPA). However, poor images are associated with miosis, corneal opacities and ocular media opacities. In addition, images with a leopard spot appearance, even though they are clear, also make RNFL defects more difficult to detect in Asian countries.¹⁰

On the other hand, thinning of the RNFL and ganglion cell inner plexiform layer (GCIPL) increases with glaucomatous progression, which can be detected by optical coherence tomography (OCT). Most OCT reports provide peripapillary RNFL and macular GCIPL sectors with both numerical data and colour-coded classification according to normative databases.^{11 12} However, there is no standardised diagnostic procedure for glaucoma using OCT. Results may vary depending on the interpretation of the ophthalmologist.¹³ Therefore, it may be possible to improve the accuracy of glaucoma diagnosis by combining fundus photography and OCT.

The number of glaucoma specialists in China is limited. In recent years, the development of artificial intelligence (AI) has made it possible to diagnose eye diseases automatically.¹⁴ In particular, diagnostic capabilities are being enhanced by machine learning (ML), including deep neural networks (also known as deep learning (DL)).¹¹ DL has seen a surge of research in the medical field in recent years because of its deeper algorithmic structure, which allows it to self-train and discover inherent rules from large amounts of unlabeled data.^{15 16} Currently, two main DL models exist, including convolutional neural network (CNN) and large-scale training of an artificial neural network. The introduction of CNN has provided a breakthrough in developing DL for classifying images and recognising patterns. CNNs are widely used as a model architecture for image classification tasks due to their effectiveness and stability. They automatically extract local spatial features from images through convolution operations, progressively constructing high-level semantic representations using multilayer structures. The parameter sharing mechanism in CNNs also improves training efficiency and reduces the risk of overfitting. Given these advantages, CNNs are widely used for computer vision tasks such as image recognition and object detection, and their effectiveness for image modelling has been proven in practice.¹⁷ Therefore, AI may be a long-term solution for glaucoma diagnosis and monitoring.¹⁸

This study was designed to establish a multimodal deep-learning algorithm to discriminate between POAG and normal subjects by combining fundus photographs, RNFL and GCIPL thickness maps, and deviation maps.

MATERIALS AND METHODS

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting or dissemination plans of our research.

Subjects

Eyes were selected between January 2018 and June 2023 from the database of subjects examined for glaucoma at the Joint Shantou International Eye Center. This study was approved by the Institutional Review Board of the Joint Shantou International Eye Center and adhered to the tenets of the Declaration of Helsinki. Informed consent was waived due to the retrospective nature of the study and anonymised database usage.

All subjects had a complete ophthalmic examination before the study, including corrected visual acuity, refraction, slit-lamp biomicroscopy, intraocular pressure measurement, standard automated perimetry (SAP; Humphrey Field Analyzer II; Carl Zeiss Meditec, Dublin, California), in addition to fundus photography (Topcon TRC-50DX, Japan) and OCT (Cirrus HD-OCT (Carl Zeiss Meditec, Oberkochen, Germany)).

Inclusion criteria were as follows: (1) all participants were aged ≥ 18 years (2) subjects were diagnosed with POAG based on fundus photography, OCT, SAP and medical history. The exclusion criteria were: (1) OCT images with poor quality (signal strength < 6 for Cirrus HD-OCT), motion artefacts and uninterpretable reports; (2) fundus photographs with poor location (such as a not fully visible optic disc); (3) the presence of a secondary cause of glaucomatous optic neuropathy (GON); (4) any retinal disease that could affect the GCIPL and RNFL; (5) a history of intraocular surgery (other than cataract surgery) or retinal laser photocoagulation; (6) any neurologic or systemic disease that could affect the retinal outcome. The diagnostic criteria for POAG are as follows^{19 20}: (1) evidence of optic nerve damage from either, or both, of the following: (a) optic disc or RNFL structural abnormalities: diffuse or focal narrowing, or notching, of the optic disc rim; progressive narrowing of the neuroretinal rim with an associated increase in cupping of the optic disc; diffuse or localised thinning of the parapapillary RNFL; optic disc haemorrhages involving the disc rim, parapapillary RNFL or lamina cribrosa; optic disc neural rim asymmetry of the two eyes consistent with loss of neural tissue; beta-zone PPA; thinning of the RNFL and/or macula on imaging. (b) Reliable and reproducible visual field examinations: fixation loss, false positive and false negative were all less than 15%; a cluster of 3 points with probabilities $< 5\%$ in at least one hemifield on the pattern deviation map (including ≥ 1 point with probability of $< 1\%$ or a cluster of two points with a probability of $< 1\%$); outside of the normal limits on glaucoma hemifield test; or a pattern SD $< 5\%$, as confirmed by at least two reliable examinations. (2) adult onset; (3) open anterior chamber angles; (4) absence of other known explanations for progressive

glaucomatous optic nerve change. Inclusion criteria for the normal group were no abnormal findings except for a clinically insignificant senile cataract on biomicroscopy and no history of any ocular disease that could have influenced the results of the OCT examinations. The visual field examinations of included normal eyes were within normal limits (fixation loss, false positive and false negative were all less than 15%). Both eyes were included if both eyes of the same subjects met the inclusion criteria.

Clinical examinations

Scans of the optic disc (optic disc cube 200×200) and the macula (macula cube 512×128) were performed using the Cirrus HD-OCT software for RNFL and GCIPL thickness measurements, respectively. Fundus photographs were taken at an angle of 45°, centred on the macula.

Image processing and dataset establishment

In the preprocessing step, all images in the dataset were down-sampled to a uniform size of 299×299 pixels, and pixel values were scaled to range from 0 to 1. Data augmentation techniques were used on the training dataset to improve image heterogeneity, avoid the possibility of overfitting and improve the generalisation and robustness of the model.

First, there is confirmation of the patient's specific diagnosis based on the gold standard for POAG diagnosis. Subsequently, imaging data (eg, fundus photographs and OCT images) corresponding to these confirmed patients is collated and annotated. The 'glaucoma group' is where all images corresponding to a patient diagnosed with glaucoma are classified. The 'normal group' is where all images corresponding to a patient diagnosed as normal are classified. The use of this method ensures clear and high-quality diagnostic labels for images used for model training or analysis.

For each subject, the following four images were used to input the CNN models: (1) GCIPL and RNFL deviation map; (2) GCIPL thickness map; (3) RNFL thickness map; (4) fundus photography. The DL model needs as many training samples as possible to achieve high accuracy. If a subject underwent OCT multiple times, images were included in this study. At the participant level, the images were randomly divided into training (64%), validation (16%) and test (20%) samples, which avoided leakage and distorted estimates of test performance by ensuring that no participant had data in both the training and test samples.

Development of DL algorithm

This multidimensional imaging model used four basic CNN models (Swin_base_patch4_window7_224_in22k, Inception ResNet V2, inception_V3, Xception). First, only one type of image (ie, fundus photography) was used to train each independent CNN model. The training dataset was input to these four CNN models, and by iterating over batches of training data, the models automatically adjusted the parameters to minimise the

loss function via a backpropagation algorithm. At the end of each training epoch, the training progress of the model was evaluated based on the accuracy of the training dataset. Additionally, various aspects of the network architecture, including the number of layers, the number of filter banks per layer, the size of the filters, the number of steps and the use of batch normalisation, were optimised by randomly exploring the hyperparameters, similar to the hyperparameter optimisation carried out for the feature-based approach. We selected significant hyperparameters and then uniformly sampled 1000 times for each training fold. The CNN model internally used the rectified linear unit (ReLU) activation function normalised at the hidden layer output to improve model stability. Simultaneously, the Adam optimiser was used for training the network. In each model, the Softmax classifier processed the input image features to output the probability that the subject was glaucomatous. The parameters with the highest AUC in the validation set were used to calculate the AUC in the test set. This process was repeated 5 times (5-fold cross-validation). Each fold randomly divides all the data into five parts and takes one as the test dataset. The remaining four parts are then equally divided into five parts; these are taken as one validation set and four parts as the training dataset. Repeat the above process five times.

Second, the outputs of these four independent CNN models for each type of image were unweighted averages to derive the prediction probabilities of (1) the 'fundus photography model', (2) the 'GCIPL and RNFL deviation map model', (3) the 'GCIPL thickness map model' and (4) the 'RNFL thickness map model' by using the model ensemble technique.

Third, the back-end mean fusion technique is used to combine the prediction results of the above ensemble models by averaging the integrated models of different images to obtain multimodal models for different image combinations: (1) 'GCIPL and RNFL thickness map model'; (2) 'fundus photography+GCIPL and RNFL deviation map model'; (3) 'fundus photography+GCIPL and RNFL thickness map model'; (4) 'GCIPL and RNFL deviation map+GCIPL and RNFL thickness map model'; (5) 'fundus photography+GCIPL and RNFL deviation map+GCIPL and RNFL thickness map model'. This process did not require any change in the model structure or training process, which can reduce the error and variance of individual models and improve the robustness and accuracy of the multimodal models. The process of developing the multimodal model is shown in figure 1.

Visualising model decisions

Gradient-weighted class activation mapping (CAM) is a gradient-based localisation technique that makes the CNN model more explicable by generating a visual explanation highlighting the class-discriminating region in the image for predicting the decision of interest.²¹ Therefore, we constructed heatmaps corresponding to the CAM over the input images, visualising the parts of the

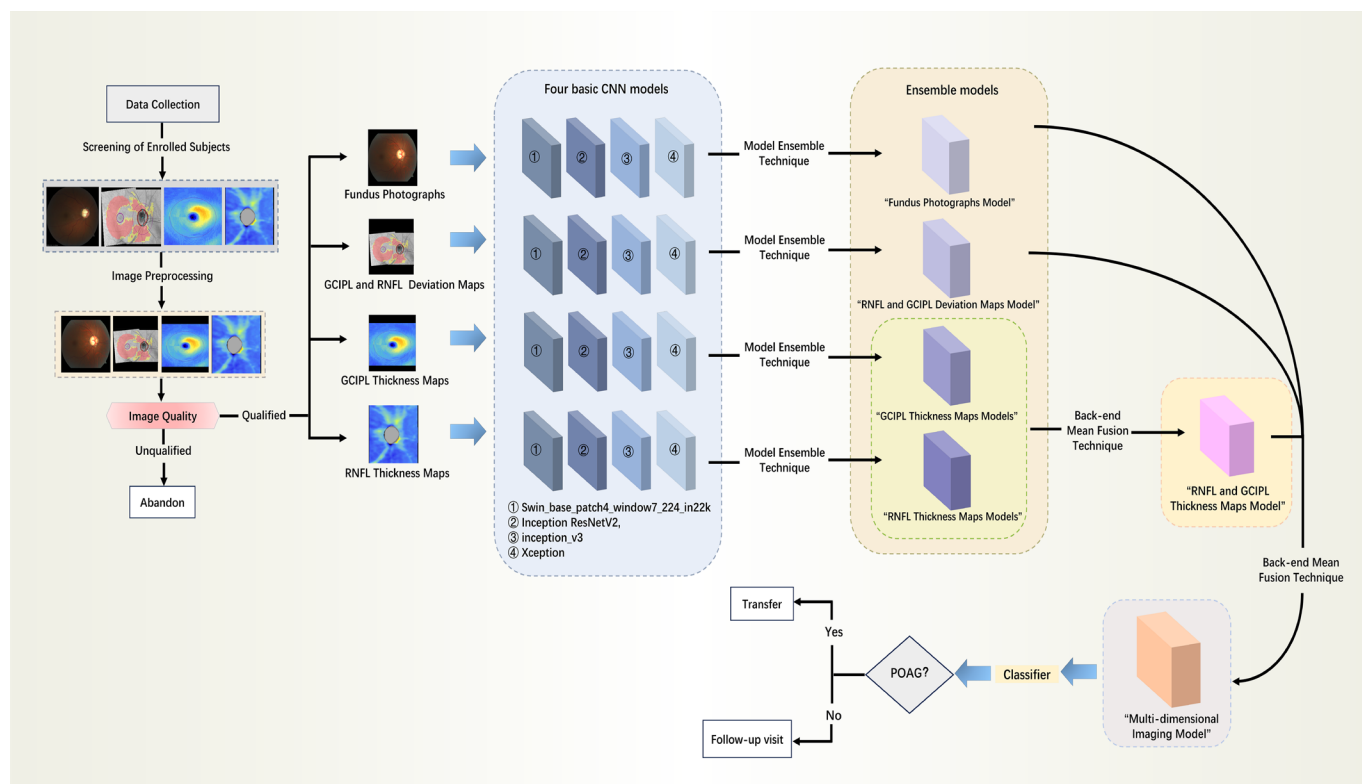


Figure 1 Development of deep learning algorithm. Each type of image was used to train each CNN model shown above. The model ensemble technique combined the outputs of these four independent CNN models for each type of image. The back-end mean fusion technique combined the prediction results of the above four ensemble models to develop the multi-dimensional imaging model. CNN, convolutional neural network; GCIPL, ganglion cell-inner plexiform layer; POAG, primary open-angle glaucoma; RNFL, retinal nerve fibre layer.

image that are most important for the classification by the multidimensional imaging model.

Comparison between CNN-based algorithm and ophthalmologists

We compared the diagnostic accuracy between our multi-dimensional imaging model and ophthalmologists. We selected three ophthalmologists at two levels (attending ophthalmologists: Dr Xiaoling Xie and Kailin Xiao; resident ophthalmologist: Dr Lipin Shi). The attending ophthalmologists had at least 5 years of clinical ophthalmological experience, while the resident ophthalmologist had 1–3 years of clinical ophthalmological experience. The ophthalmologists were shown the images independently and asked to assign one of two labels to each set of images: non-glaucomatous or glaucomatous.

Comparison of POAG associated with myopia and non-myopic POAG

Patients with POAG in the test dataset were divided into myopic and non-myopic groups together with healthy subjects and entered into the models. The myopic group was further divided into three groups based on equivalent spherical lenses for mild, moderate and high myopia: mild group ($-3D < SE < -0.5D$), moderate group ($-6D < SE \leq -3D$) and high group ($SE \leq -6D$). Data

were analysed, and differences between the groups were compared using statistical software.

Statistical analysis

We calculated the overall accuracy, sensitivity, specificity and area under the receiver operator characteristic curve (AUC) metrics with a 95% CI to assess the performance of the DL algorithms. Between normal and glaucoma groups, we used an unpaired t-test for numerical data and a χ^2 test for categorical data. The DeLong test was used to calculate p values for comparing AUCs between groups. Sensitivity and specificity were calculated following the optimal cut-off point, which was the maximum of the Youden index. All statistical analyses were performed with SPSS (V.19.0.0; SPSS, Chicago, Illinois) and MedCalc (V.20.022; MedCalc, Ostend, Belgium) for statistics. P values < 0.05 were considered statistically significant. Data are presented as mean $\pm$ SD.

RESULTS

Characteristics of the study participants

The baseline characteristics of the study participants are summarised in online supplemental table S1. 4216 pairs of fundus photographs and OCT images (deviation and thickness maps) were collected from 546 patients. We used an unpaired t-test for numerical data and a χ^2 test

for categorical data to compare the statistical difference between normal and glaucoma groups. It was observed that there was no significant difference between the left-eye and right-eye ratio ($p=0.691$). However, age ($p<0.001$), gender ($p=0.040$), RNFL thickness ($P<0.001$), GCIPL thickness ($P<0.001$) and mean deviation of the visual field ($P<0.001$) showed significant statistical differences.

Performance of DL algorithms

We evaluated the diagnostic performance of the algorithms for differentiating glaucoma from the normal group based on fundus photographs, RNFL and GCIPL deviation maps, RNFL and GCIPL thickness maps, and fundus photograph/OCT pairs separately, as shown in online supplemental table S2).

In the training dataset, 'Fundus photographs+RNFL and GCIPL deviation+thickness maps model' got a higher AUC of 0.996 (95% CI 0.993 to 0.997) than 'Fundus photographs model' (AUC, 0.992 (95% CI 0.989 to 0.995); $p<0.05$), 'RNFL and GCIPL deviation maps model' (AUC, 0.985 (95% CI 0.980 to 0.988); $p<0.001$) and 'RNFL and GCIPL thickness maps model' (AUC, 0.982 (95% CI 0.976 to 0.986); $p<0.001$). In the validation dataset, the multidimensional imaging model (AUC, 0.977 (95% CI 0.964 to 0.986)) performed better than the 'Fundus photographs model' (AUC, 0.932 (95% CI 0.912 to 0.948); $p<0.01$), 'RNFL and GCIPL deviation maps model' (AUC, 0.950 (95% CI 0.932 to 0.964); $p<0.005$) and 'RNFL and GCIPL thickness maps model' (AUC, 0.973 (95% CI 0.959 to 0.983)). In the test dataset, the multidimensional imaging model (AUC, 0.970 (95% CI 0.958 to 0.979)) outperformed 'Fundus photographs model' (AUC, 0.945 (95% CI 0.929 to 0.958); $p<0.05$), 'RNFL and GCIPL deviation maps model' (AUC, 0.931 (95% CI 0.915 to 0.946); $p<0.001$) and 'RNFL and GCIPL thickness maps model' (AUC, 0.958 (95% CI 0.944 to 0.969); online supplemental figure S1).

Comparison of the multidimensional imaging model with the glaucoma ophthalmologists

We chose one fold of the test dataset to compare the multidimensional imaging model with the glaucoma ophthalmologists. The multidimensional imaging model (AUC, 0.969 (95% CI 0.934 to 0.988)) performed better than three ophthalmologists (attending ophthalmologist 1: AUC, 0.847; attending ophthalmologist 2: AUC, 0.842; resident ophthalmologist: AUC, 0.755; $p<0.001$ for both, online supplemental table S3 and figure S2).

Visualisation of the multidimensional imaging model

Examples of eyes with GON detected by the 'Fundus photo+RNFL and GCIPL deviation+thickness maps model' and corresponding heat maps are shown in figure 2. It was able to localise the area of pathologic characteristics on RNFL and GCIPL deviation maps, GCIPL thickness maps, RNFL thickness maps and fundus photographs.

Analysis of POAG associated with myopia and non-myopic POAG

In this study, a total of 827 myopic eyes were collected: (1) 288 eyes in the mild myopic group (including 133 eyes in the normal group, with a total of 532 paired fundus photographs and OCT images, and 155 eyes in the glaucoma group, with a total of 620 paired fundus photographs and OCT images); (2) 264 eyes in the moderate myopic group (including 179 eyes in the normal group, with a total of 716 paired fundus photographs and OCT images, and 85 eyes in the glaucoma group, with a total of 340 paired fundus photographs and OCT images); (3) 275 eyes in the high myopic group (including 171 eyes in the normal group, with a total of 684 paired fundus photographs and OCT images, and 104 eyes in the glaucoma group, with a total of 416 paired fundus photographs and

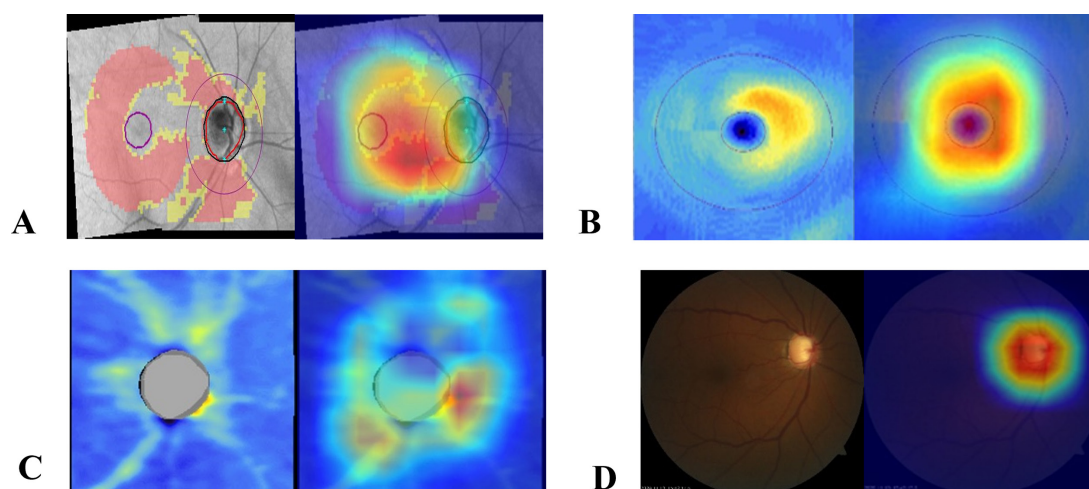


Figure 2 Class activation maps (heatmaps) of the multi-dimensional imaging model. (A) Heatmaps from RNFL and GCIPL deviation maps. (B) Heatmaps from GCIPL thickness maps. (C) Heatmaps from RNFL thickness maps. (D) Heatmaps from fundus photographs. GCIPL, ganglion cell-inner plexiform layer; RNFL, retinal nerve fibre layer.

OCT images). Baseline data in the myopia subgroup are shown in online supplemental table S4.

We compared POAG in combination with myopia and non-myopic POAG in the models (in table 1, online supplemental figure S1). We found that the overall AUCs were all lower in the myopic group than in the non-myopic group. However, the AUC of the multidimensional imaging model in POAG with myopia could reach 0.966 (95% CI 0.951 to 0.977).

In the myopia group, we compared the diagnostic performance of mild myopia, moderate myopia and high myopia datasets for POAG in each of the models (in online supplemental table S5 and figure S1). We found that the overall AUC for lesion detection by each model was: low myopia group > moderate myopia group > high myopia group.

Additionally, patients with POAG were classified according to their mean visual field defect (mean deviation, MD) using the Hodapp-Parrish-Anderson criteria into the following groups: early defect group (MD $\geq$ -6 dB, 239 eyes), moderate defect group (-12 dB $\leq$ MD < -6 dB, 79 eyes) and severe defect group (MD < -12 dB, 191 eyes). In the POAG group, we compared the diagnostic performance of early defect, moderate defect and severe defect datasets in each of the models (in online supplemental table S6). We found that the multidimensional imaging model has the best performance in each POAG group. The overall AUC for lesion detection by each model was severe defect group > moderate defect group > early defect group.

A total of 216 glaucomatous eyes under the age of 50 (61 of which were females) and 293 glaucomatous eyes over the age of 50 (90 of which were females). Online supplemental table S7 shows the results of the stratified analysis in the test dataset after adjusting for confounding factors. AUC of the multidimensional imaging model was not significantly affected by gender within the same age group (ie, <50 years or $\geq$ 50 years) ($p > 0.05$). At the same time, within the same gender group, age had no significant effect on the AUC of the multidimensional imaging model ($p > 0.05$).

DISCUSSION

In the present study, we have developed and evaluated a multidimensional imaging model combining fundus photographs with RNFL and GCIPL thickness maps and deviation maps for the POAG diagnostic performance. The current results demonstrated that this multidimensional imaging model could identify GON with high accuracy, sensitivity, specificity and AUC. In addition, it could outperform traditional unimodal models and clinical ophthalmologists. The multimodal model also performed well in myopia combined with POAG. In addition, we constructed heatmaps corresponding to the gradient-weighted CAM over the input images. This helped to localise the area of pathological features on the images.

In clinical practice, multiple examinations such as VF, OCT and fundus photography are always combined to diagnose glaucoma.²² In addition, detecting structural damage in glaucoma is more relevant to glaucoma diagnosis as it occurs earlier than functional damage and has been extensively studied.⁶ Previous studies have used traditional approaches to develop single ML algorithms or DL algorithms.^{2 11 23 24} However, it has been shown by studies that myopia, especially moderate to high myopia, is considered to be an independent risk factor for glaucoma, with the risk of developing glaucoma being increased by 2–3 times.²⁵ Myopia has been confirmed as a common ophthalmological condition. Epidemiological survey results indicate that, by 2050, the global number of patients with myopia will increase to 4.758 billion, with 938 million of these having high myopia, accounting for approximately 9.8% of the worldwide population.²⁶ Changes in retinal structure associated with myopia, such as optic disc deformation, tilting and rotation, retinal choroidal atrophy and posterior scleral staphyloma, are currently making the screening and diagnosis of POAG more challenging. These changes often make the typical retinal manifestations of POAG less noticeable, such as loss of the optic disc rim, increased cup-to-disc ratio, rim notching and optic nerve fibre layer atrophy. The RNFL is now commonly used to diagnose glaucoma, but its clinical use, especially in myopia with glaucoma, is limited by the influence of age, disc size, disc tilt, vessel position²⁷ and axial length, which refers to the distance from the anterior surface of the cornea to the retinal pigment epithelium or foveal region of the retina along the optical axis of the eye. In this study, we also combined fundus photography, which can directly detect changes in the vertical cup-to-disc ratio and retinal vascularity. It can correct the effects of vascularity and optic disc on OCT measurements, thus improving the diagnostic accuracy of myopia combined with glaucoma. However, fundus photography evaluates the retina and optic nerves in a two-dimensional plane. OCT displays the results in three dimensions to find RNFL defects, often preceding signs such as optic disc cupping enlargement and visual field loss.¹⁰ Taken together, adding OCT to fundus photography in DL algorithms could improve the accuracy of glaucoma diagnosis.

Watanabe *et al*¹⁰ evaluated the accuracy of glaucoma diagnosis using fundus photography alone and fundus photography combined with OCT images in 1006 eyes of 503 patients who underwent a healthy follow-up examination. Their research showed that fundus photography alone had a low sensitivity of 55% for detecting glaucoma, whereas adding OCT improved the sensitivity to 80%. Li *et al*²³ developed a deep-learning algorithm to detect GON from fundus photographs. They found that image quality problems could have negatively affected model predictions, resulting in model errors and low-confidence predictions. Lee *et al*¹¹ established a DL algorithm based on the GCIPL thickness map, GCIPL deviation map, RNFL thickness map, and RNFL

Table 1 Comparison of POAG with myopia and non-myopic POAG in the test dataset

Dataset/model	AUC (95% CI)	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	F1 score	P value
Myopic group						
(1) Fundus photographs model	0.938 (0.919 to 0.953)	0.878 (0.856 to 0.900)	0.875 (0.852 to 0.898)	0.880 (0.858 to 0.902)	0.856	0.006
(2) RNFL and GCIPL deviation maps model	0.935 (0.916 to 0.951)	0.868 (0.845 to 0.891)	0.863 (0.840 to 0.887)	0.872 (0.849 to 0.894)	0.845	0.005
(3) RNFL and GCIPL thickness maps model	0.958 (0.942 to 0.971)	0.898 (0.878 to 0.919)	0.875 (0.852 to 0.898)	0.915 (0.896 to 0.934)	0.878	0.406
(4) RNFL and GCIPL deviation+thickness maps model	0.954 (0.937 to 0.967)	0.901 (0.880 to 0.921)	0.875 (0.852 to 0.898)	0.919 (0.901 to 0.938)	0.880	0.207
(5) Fundus photographs+RNFL and GCIPL deviation maps model	0.961 (0.946 to 0.973)	0.897 (0.877 to 0.918)	0.901 (0.881 to 0.922)	0.894 (0.873 to 0.915)	0.879	0.606
(6) Fundus photographs+RNFL and GCIPL thickness maps model	0.967 (0.952 to 0.978)	0.903 (0.883 to 0.923)	0.898 (0.878 to 0.919)	0.907 (0.887 to 0.927)	0.885	0.911
(7) Fundus photographs+RNFL and GCIPL deviation+thickness maps model	0.966 (0.951 to 0.977)	0.908 (0.888 to 0.928)	0.910 (0.890 to 0.929)	0.907 (0.887 to 0.927)	0.892	–
Non-myopic group						
(1) Fundus photographs model	0.940 (0.900 to 0.967)	0.924 (0.889 to 0.959)	0.994 (0.984 to 1.000)	0.714 (0.655 to 0.774)	0.951	0.157
(2) RNFL and GCIPL deviation maps model	0.895 (0.848 to 0.932)	0.830 (0.780 to 0.879)	0.862 (0.817 to 0.908)	0.732 (0.674 to 0.790)	0.883	0.002
(3) RNFL and GCIPL thickness maps model	0.939 (0.900 to 0.967)	0.874 (0.831 to 0.918)	0.886 (0.845 to 0.928)	0.839 (0.791 to 0.887)	0.914	0.084
(4) RNFL and GCIPL deviation+thickness maps model	0.925 (0.883 to 0.956)	0.879 (0.836 to 0.922)	0.904 (0.866 to 0.943)	0.804 (0.751 to 0.856)	0.918	0.023
(5) Fundus photographs+RNFL and GCIPL deviation maps model	0.971 (0.939 to 0.989)	0.910 (0.873 to 0.948)	0.976 (0.956 to 0.996)	0.714 (0.655 to 0.774)	0.942	0.974
(6) Fundus photographs+RNFL and GCIPL thickness maps model	0.980 (0.951 to 0.994)	0.901 (0.862 to 0.94)	0.976 (0.956 to 0.996)	0.679 (0.617 to 0.740)	0.937	0.469
(7) Fundus photographs+RNFL and GCIPL deviation+thickness maps model	0.971 (0.940 to 0.989)	0.901 (0.862 to 0.94)	0.958 (0.932 to 0.984)	0.732 (0.674 to 0.790)	0.936	–
Data are presented with a 95% CI. The DeLong test was used to calculate p values for the comparison of areas under the receiver operator characteristic curve between groups.						
AUC, area under the receiver operating characteristic curve; GCIPL, ganglion cell-inner plexiform layer; POAG, primary open-angle glaucoma; RNFL, retinal nerve fibre layer.						

deviation map. They found that artefacts may cause the misclassification. Therefore, to reduce the model error of unimodal models, we developed a multidimensional imaging model that combined fundus photographs with the RNFL and GCIPL thickness maps and deviation maps, which may provide more pathological features on the images. The current study also found consistent results: the multidimensional imaging model outperformed the unimodal DL model. In the test dataset, the multidimensional imaging model obtained an AUC of 0.970, while the AUC of the 'fundus photographs model' was 0.945, the AUC of the 'RNFL and GCIPL deviation maps model' was 0.931, and the AUC of 'RNFL and GCIPL thickness maps model' was 0.958.

Previous studies have shown that combining CNN-based DL algorithms with fundus photography has improved the diagnostic accuracy of glaucoma. DL algorithms can directly extract structural features related to glaucomatous lesions, such as cup-to-disc ratio, optic disc haemorrhages, etc, from fundus photographs. Ting *et al*²⁸ collected 125 189 fundus photographs to develop a CNN-based DL algorithm for glaucoma detection based on cup-to-disc ratio, RNFL defects, etc. They obtained an AUC of 0.942, a sensitivity of 0.964 and a specificity of 0.872. Christopher *et al*²⁹ collected 14 822 fundus photographs with a resolution of 224×224 and constructed a deep-learning algorithm based on ResNet50. This algorithm has an AUC of 0.910, a sensitivity of 0.850 and a specificity of 0.800 for detecting GON. Currently, most diagnostic studies of POAG have focused on analysing structural changes. Hirasawa *et al*³⁰ used SD-OCT to detect peripapillary RNFL changes in healthy subjects and patients with early POAG. The results showed that peripapillary RNFL changes may be an effective method to detect early POAG. Moreover, there have been many recent studies on the detection of glaucomatous optic neuropathy using a combination of OCT images and DL. Asaoka *et al*² constructed a CNN model with an AUC of 0.937 using macular RNFL and GCIPL thicknesses in 4317 OCT images as input parameters. It was shown that the DL model significantly improves the diagnostic performance of the OCT. Muhammad *et al*³¹ collected OCT images to obtain retinal ganglion cell plus inner plexiform layer (RGC+) thickness maps, RNFL thickness maps and corresponding probability maps and used them with CNN to develop a model with an AUC of 0.950. In addition, these studies did not evaluate the diagnostic performance of the model in a population of people with myopia. In the present study, we also used OCT for RNFL detection and combined it with fundus photography to obtain the RNFL, disc and vessel characteristics, which complement one another and improve the diagnostic accuracy for different stages of POAG, especially in the early stages. We constructed a multidimensional imaging model by combining fundus photography with RNFL and GCIPL deviation maps and thickness maps, which achieved an AUC of 0.970, a sensitivity of 0.926 and a specificity of 0.889. There was an improvement in diagnostic accuracy

over the 'fundus photographs model' (AUC, 0.945), the 'RNFL and GCIPL deviation maps model' (AUC, 0.931) and the 'RNFL and GCIPL thickness maps model' (AUC, 0.958). Meanwhile, the diagnostic accuracy of the multidimensional imaging model in myopia populations was explored, which maintained a high diagnostic AUC of 0.966. It also found that AUC could reach 0.958 in high myopia populations. Both in the myopic population and even in the high-myopic population, the multidimensional imaging outperforms the unimodal models. In the training, validation and test datasets, we found that the 'fundus photographs+RNFL and GCIPL thickness maps model' (AUC 0.972–0.995) performed almost identically to the multidimensional imaging model (AUC 0.970–0.996). Further comparison of the performance of the unimodal models reveals that the diagnostic efficacy of the 'RNFL and GCIPL thickness maps model' (AUC 0.958–0.982) is marginally superior to that of the 'RNFL and GCIPL deviation maps model' (AUC 0.931–0.985). However, in the POAG group, a comparison was made of the diagnostic performance of the early defect group, moderate defect group and severe defect group in each of the models. The 'RNFL and GCIPL deviation maps model' had a markedly pronounced advantage, with its AUC values (0.919–0.997) consistently exceeding those of the 'RNFL and GCIPL thickness maps model' (AUC 0.871–0.994). In particular, the 'RNFL and GCIPL deviation maps model' in the early defect group achieved an AUC of 0.919, which is significantly higher than the 'RNFL and GCIPL thickness maps model' (AUC 0.871). These results demonstrate that integrating deviation maps into the multidimensional imaging model effectively improves the identification of early-stage glaucoma, providing superior technical support for early disease screening. Additionally, by analysing the false-negative results, we found that myopic features such as optic disc paracentral atrophy and optic disc tilt were the leading causes of the false-negative classification of the high myopia group in the current study. Similarly, Li *et al*³² found that optic disc features secondary to pathological myopia or high myopia (42.6% of the total number of misclassifications) were the most common cause of misclassification in their study of a DL algorithm for glaucoma based on fundus photographs. Ran *et al*⁵ also mentioned that myopic features such as paracentral optic nerve atrophy, optic nerve tilt and leopard-like fundus were identified as one of the causes of false negatives and positives.

Despite the results of the current study on glaucoma diagnosis using multidimensional imaging, some limitations should be considered. First, the study subjects were all selected from the clinical population, which may have resulted in a higher prevalence of glaucoma than in the general population, limiting the generalisability of the results. To improve the generalisability of the model, future studies should consider the application of this multidimensional imaging model to a broader range of populations, including health screening and community screening. Second, the glaucoma patients in this study

were generally older than the non-glaucoma subjects, which may have affected the results. To more accurately assess the impact of age factors on glaucoma diagnosis, future studies are recommended to increase the sample size and achieve a balanced age distribution or to apply appropriate statistical controls to the age variable during analysis. In addition, the sample in the current study lacked racial diversity, which may have limitations in the applicability of the findings to racially diverse populations. Additionally, this study has not yet validated the model using external independent datasets, which is an important limitation of the current work. Future work will focus on introducing multicentre or cross-platform data in order to evaluate the model's robustness and potential for practical application more comprehensively. Therefore, future studies should recruit more participants from diverse racial and cultural backgrounds to ensure that the findings better reflect and apply to the characteristics of diverse populations worldwide. Finally, a relatively single brand of fundus photography and OCT equipment was used in this study. This may have impacted the adaptability and accuracy of the multidimensional imaging model under different equipment conditions. Future studies should explore and evaluate imaging data from various brands of devices.

In summary, our study has established a high-performance multidimensional imaging model that combines fundus photographs with the RNFL and GCIPL thickness maps and deviation maps. It could outperform traditional single-modal algorithms and clinical ophthalmologists. It also improves the diagnostic accuracy of myopia combined with POAG for various levels of myopia. Therefore, it has the potential to help ophthalmologists diagnose patients with POAG during POAG screening.

Acknowledgements We used the STROBE cross-sectional checklist when writing our report. We would like to express our sincere gratitude to Dr Xiaoling Xie and Kailin Xiao for their invaluable help in reviewing the images for this study.

Contributors GW and JQ: contributed to conception, design, analysis and interpretation of data. Revised paper for important intellectual content and final approval of the version submitted for publication. JL: contributed to developing algorithms. LS: contributed to image collection and diagnosis. KQ and CH: contributed to method guidance. JQ is the guarantor.

Funding This study was supported by the Shantou Municipal Science and Technology Project (Grant Number: 190917155269927) and Special Fund for Central Guiding Local Science and Technology Development Project (Grant Number: STKJ2024064).

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This retrospective study has been approved by the Human Medical Ethics Committee of the Joint Shantou International Eye Center of Shantou University (EC 20190911(4)-P15). Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer-reviewed.

Data availability statement Data may be obtained from a third party and are not publicly available. Data are not published as open data.

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